Factors, Multiples & Primes

Year 4 · **Mark Scheme**

Lesson Worksheet — Answers

Easy / Standard

1. Not prime — it has factors other than 1 and itself (e.g. 2)

2. 1, 2, 5, 10

3. A, B and D (16, 24 and 32 are multiples of 8; 28 and 36 are not)

4. No (33 ÷ 5 = 6.60, not exact)

5. 49

6. Not prime — it has factors other than 1 and itself (e.g. 3)

7. C — 5 (21's factors are 1, 3, 7 and 21)

8. 7, 14, 21, 28, 35

9. No (34 ÷ 10 = 3.40, not exact)

10. D — 25

11. Not prime — it has factors other than 1 and itself (e.g. 2)

12. 1, 2, 4, 8

13. C — 45

14. No (31 ÷ 4 = 7.75, not exact)

15. C — 16

Standard

16. 2, 3, 5, 7, 11, 13, 17, 19, 23

17. C — Yes, $9 = 3^2$, so $\sqrt{9} = 3$

18. 1, 2, 3, 6

19. 30

20. 1×30, 2×15, 3×10, 5×6

21. A and C (41 and 43 are prime; 42, 45 and 51 all have factors other than 1 and themselves)

22. Yes, $49 = 7^2$, so $\sqrt{49} = 7$

23. 1, 2, 4

24. B — 4

25. 1×36, 2×18, 3×12, 4×9, 6×6

26. 2, 3, 5, 7, 11, 13, 17, 19

27. C — Yes, $81 = 9^2$, so $\sqrt{81} = 9$

28. 1, 2, 5, 10

29. D — 30

30. 1×48, 2×24, 3×16, 4×12, 6×8

Standard / Hard

31. Square number — $16 = 4^2$

32. D — 12 (LCM of 4 and 6 is 12)

33. 3×5=15 (odd)

34. 8 factors: 1, 2, 3, 4, 6, 8, 12, 24

35. B — 6

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36. Square number — $25 = 5^2$

37. A, C and D (4, 6 and 9 are factors of 36; 5 and 10 are not)

38. LCM of 5 and 2 = 10 minutes

39. B — 35, odd

40. 8 factors: 1, 2, 3, 5, 6, 10, 15, 30

41. Common factors: 1, 5. HCF = 5

42. C — Neither ($27 = 3^3$, which is not prime and not a perfect square)

43. LCM of 6 and 9 = 18 minutes

44. A, C and D (5, 8 and 10 are factors of 40; 6 and 12 are not)

45. Common factors: 1, 2, 4. HCF = 4

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Homework Worksheet — Answers

Easy / Standard

1. No (18 ÷ 5 = 3.60, not exact)

2. A, C and E (4, 10 and 18 are multiples of 2; 7 and 15 are odd)

3. Prime — only divisible by 1 and 5

4. 1, 3, 7, 21

5. 49

6. C — No (36 ÷ 5 = 7.20, not exact)

7. 3, 6, 9, 12, 15

8. Prime — only divisible by 1 and 3

9. C — 9 (24's factors are 1, 2, 3, 4, 6, 8, 12, 24)

10. 36

11. No (26 ÷ 10 = 2.60, not exact)

12. C — 24

13. Not prime — it has factors other than 1 and itself (e.g. 2)

14. A, B, D and E (1, 2, 5 and 10 are factors of 10; 3 is not)

15. 81

Standard

16. 12

17. A, B and D (4×10=40, 5×8=40, 2×20=40; C and E do not equal 40)

18. 1, 2, 5, 10

19. Yes, $100 = 10^2$, so $\sqrt{100} = 10$

20. 2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31

21. C — 20

22. 1×30, 2×15, 3×10, 5×6

23. 1, 2, 4, 8

24. C — Yes, $64 = 8^2$, so $\sqrt{64} = 8$

25. 2, 3, 5, 7, 11, 13, 17, 19, 23

26. 10

27. D — 3 × 9 = 27, not 24

28. 1, 5

29. B — Yes, $49 = 7^2$, so $\sqrt{49} = 7$

30. 2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47, 53

Standard / Advanced

31. 33 = 3×11, not prime — smallest factor other than 1 is 3

32. Yes, $\sqrt{36} = 6$; + 4 = 10

33. C — 12 (HCF of 8 and 12 is 4; 4×3=12)

34. LCM of 3 and 4 = 12; + 2 = 14

35. 2, 4, 10, 20

36. $36 = 6^2$, one rectangle option is 4 rows of 9 (or 3 rows of 12, or 2 rows of 18). It is a square number because 6×6 = 36

37. C — 5 (HCF of 10 and 15 is 5; each bag gets 2 croissants and 3 muffins)

38. LCM of 2 and 3 = 6 seconds

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39. Primes up to 12: 2, 3, 5, 7, 11 (5 numbers); sum = 28

40. A, B and D (2×12=24, 3×8=24, 4×6=24; C (5×5=25) and E (7×3=21) do not multiply to 24)

41. HCF of 18 and 12 = 6 bags; each bag gets 3 croissants and 2 muffins

42. D — 12 (LCM of 4 and 6 is 12)

43. $25 = 5^2$, one rectangle option is 1 row of 25. It is a square number because 5×5 = 25

44. Primes up to 15: 2, 3, 5, 7, 11, 13 (6 numbers); sum = 41

45. A, B and D (2×10=20, 4×5=20, 5×4=20 — B and D describe the same pair in a different order; C (3×7=21) and E (6×3=18) do not multiply to 20)

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